1 MATERIAL AND FRACTURE TESTS

1.1 Determination of tensile strength from a three point bending test (TPBB)

Keywords: plain concrete, softening, fracture localization, prescribed-force/prescribed-displacement boundary condition, Newton-Raphson method, rotating/fixed crack model

Input files: TPBB\TPBBF.cc2 (fixed crack)
TPBB\TPBBR.cc2 (rotated crack)

1.1.1 Introduction

Tensile strength of concrete may be indirectly determined from the result of a three point bending test on an unreinforced beam. The CEB-FIP MC90 model code [1] gives a formula, which relates the tensile strength to the measured flexural strength

$$f_{ct,fl} = f_{ctm} \frac{1 + \alpha_{fl} \left(\frac{h_b}{h_o} \right)^{0.7}}{\alpha_{fl} \left(\frac{h_b}{h_o} \right)^{0.7}} \tag{1.1}$$

Here

$f_{ct,fl}$	flexural strength (MPa)
f_{ctm}	tensile strength (MPa)
h_b	beam depth (mm)
h_o	100 (mm)
α_{fl}	coefficient, which depends on brittleness of concrete (value recommended by CEB-FIP MC90 is 1.5)

In this example, we verify validity of the above formula by simulating a three point bending experiment on a beam with known properties. The computed flexural strength is then compared to that obtained with Eq. (1.1).

1.1.2 Comments on FE model preparation

The analyzed beam geometry and support and loading conditions are shown in Figure 1-1. The input data for ATENA program are prepared through the ATENA 2-D graphical user interface (GUI) preprocessing window. Following are notes for each data block.

1.1.2.1 Materials

Concrete is modeled employing the ‘Sbeta Material’ model. Relevant material

properties used in the analysis are given in Table 1.1-1. The cubic compressive strength of 40 MPa is assumed, and the remaining parameters are left at their default values suggested by the ATENA program. The analysis is performed with both rotating and fixed crack model.

1.1.2.2 Topology

When specifying the problem topology, we place joints not only at the beam corners but also at the locations of supports and point loads to facilitate prescription of boundary conditions. The problem is modeled by a single macroelement, which is discretized by CCQ10SBeta type elements. To ensure good accuracy of the analysis, there should be at least 10 elements along the beam height. To this end, we specify element size 0.02 m.

1.1.2.3 Loads and supports and Solution parameters

Since the beam is unreinforced, we may expect that failure occurs at relatively low levels of load and reactions in supports. Thus, we may directly specify concentrated supports and loads, without a danger that these would cause local compressive crushing.

Our objective is to trace the load-displacement curve up to the post-failure softening regime. To this end, we may prescribe loading in terms of forces and use the arc-length method, which automatically changes the sign of load increment once a peak is attained. Alternatively, we may control displacement. Since displacement will keep on increasing even after the beam fails, we can employ either the Newton-Raphson method or the arc-length method. Due to its better stability, we opt for the Newton-Raphson solution method. (Note that, for large or brittle beams, a snap-back behavior may occur, in which case both displacement and force increments at the loading point change their sign upon failure. In such a situation, the arc-length solution is the only applicable method.)

To speed up convergence of the solution, we employ the full Newton-Raphson method, with tangent stiffness updated in each iteration. To automatically adjust the speed of analysis according to the nonlinearity of the response, we utilize the line search method. For this purpose we have to create a new set of ‘Solution Parameters’ (named N-R w/ LS), in which we set the appropriate options and parameters as summarized in Table 1.1-2.

1.1.2.4 Analysis steps

The load at which first cracking takes place can be estimated from the analytical elastic solution as 11.2 kN. Since the beam’s response up to this load is certainly elastic (and stable), we apply the first loading increment by prescribing force of 10 kN (load case LC2). In the following steps, we prescribe displacement increments at the loading point (load case LC3). The increment size is 0.01 mm. To accurately estimate the peak load (flexural strength), the increment size is further refined to 0.0025 mm around the peak.

1.1.2.5 Monitoring points

To facilitate the retrieval of load-displacement curves, we set up three monitoring points in the vicinity of the loading point. The first point monitors deflection, the second the applied force (associated with LC2) and the third the reaction

force at the load point (associated with LC3).

1.1.3 Results

The computed load-deflection curves obtained with the rotating and fixed crack model are shown in Figure 1-1. These data are retrieved by printing out the values at monitoring points. Note that the applied force monitored by point no. 2 changes only during the first step, when the loading is applied directly as the concentrated force. In the remaining steps, the loading is applied by prescribing displacement at the load-point, thus the additional loading force is obtained as the reaction in this point. The total applied load is then obtained as the sum of the force applied in step 1 and the reaction from the remaining steps.

Figure 1-2 shows, that the overall responses computed with the rotating and fixed crack models show almost no difference up to the tail of the post-peak part of the load-displacement curve. Crack patterns also slightly differ, especially in the final stage.

The flexural strength estimated by Eq. (1.1) with α_{fl}=1.5 and f_{ctm}=2.807 MPa, is 3.959 MPa. The computed peak load is 15.180 kN with the rotating crack model and 14.951 kN with the fixed crack model, which corresponds to flexural strengths of 3.795 MPa and 3.738 MPa, respectively. The computed results differ from the estimated strengths by 4.1% and 5.6%, respectively.

$f_{ct,fl}$ from Eq. (1.1)	$f_{ct,fl}$ computed (rotating c.)	$f_{ct,fl}$ computed (fixed c.)
3.959 (MPa)	3.795 (MPa)	3.738 (MPa)

Closer inspection of the results by displaying cracks at individual steps of analysis in ATENA post-processing GUI also reveals that initially numerous vertical cracks form at the bottom of the beam. However, later cracking localizes and the beam fails by a single crack. Such a behavior is consistent with that observed in experiments.

1.1.4 References

[1] FIB, Structural Concrete, Textbook on Behaviour, Design and Performance. Updated Knowledge of the CEB/FIB Model Code 1990, Vol. 1, FIB, 1999

Table 1.1-1 Material properties

Material type		SBETA material	
Elastic modulus	E_c	34.03	GPa
Poisson's ratio	ν	0.200	-
Compressive strength	f_c	34.000	MPa
Tensile strength	f_t	2.807	MPa
Type of tension softening		Exponential	
Fracture energy	G_f	70.18	N/m
Crack model		Fixed/Rotating	

Table 1.1-2 Solution parameters

Solution method	Newton-Raphson	
Stiffness/update	Tangent/each iteration	
Number of iterations	50	
Error tolerance	0.010	
Line search	on, with iterations	

Table 1.1-3 Finite element mesh

Finite element type	Quadrilateral, CCQ10SBeta	
Element shape smoothing	on	
Optimization	Sloan	

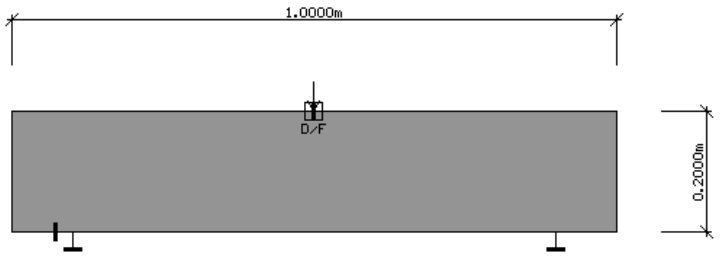


Figure 1-1: Geometry and boundary conditions of three-point bending beam

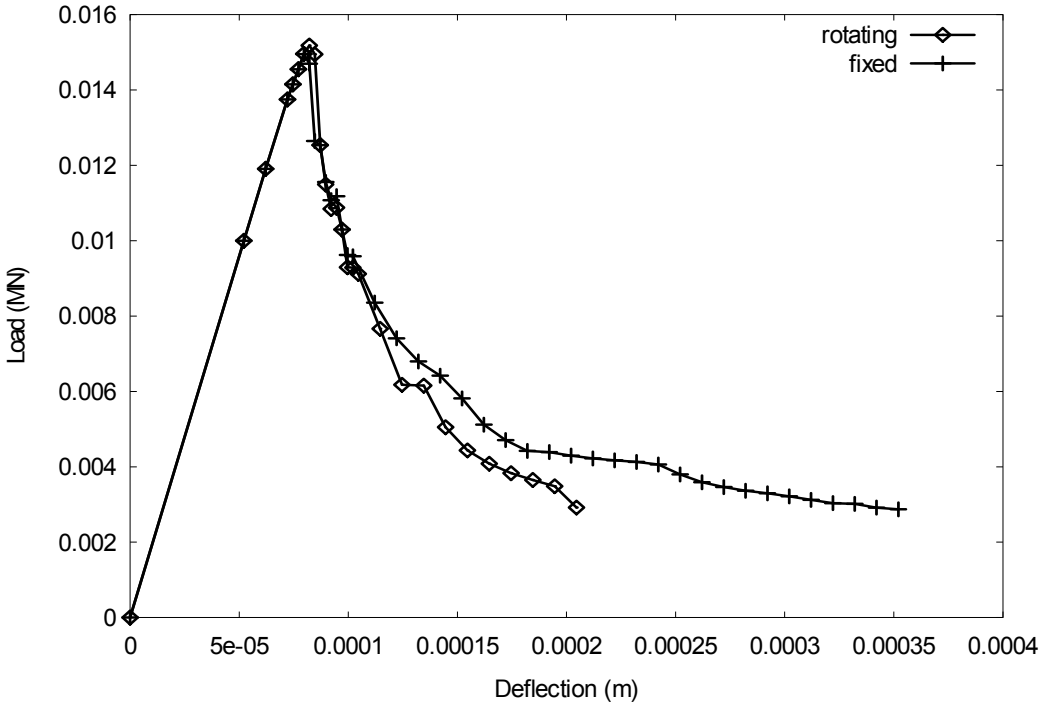


Figure 1-2: Computed load-displacement curve for a beam under three-point bending.